ASSIGNMENTS

Class-8

The Four Fundamental Subspaces

1. Describe (geometrically) the column space and row space of the following matrices:

i) $\begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix}$ iii) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

iv) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ v) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ v) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

2. Find the basis of the N(A^T) of the matrix $A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix}$

3. Find the basis for the subspace S spanned by all the solutions of $x + y + z - t = 0$

4. Find the three independent vectors on the plane $4y - 2x + 3z - t = 0$. Why not four? This plane is the null space of what matrix?